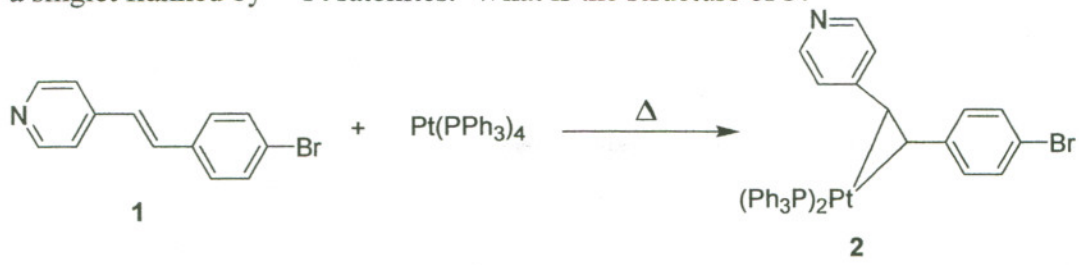
Organic Chemistry Cume – Sept 30, 2005

1) In a recent paper (*JACS*, **2005**, 9322), the van der Boom group described the conversions outlined in equations 1 and 2. Heating compound **1** in the presence of $Pt(PPh_3)_4$ led to quantitative formation of the olefin adduct **2**. Compound **2** was isolated and structurally characterized. Compound **2** can then be further heated to quantitatively provide a new Pt(II) complex **3**. The ^{31}P-proton coupled NMR spectrum of **3** consists of a singlet flanked by ^{195}Pt satellites. What is the structure of **3**?



(1)

(2)

2) Kinetic data for the reaction outlined in eq 2 are given below:

a) The reaction **2** → **3**, for which intermediates were not observed by NMR spectroscopy, follows first order kinetics in complex **2**.

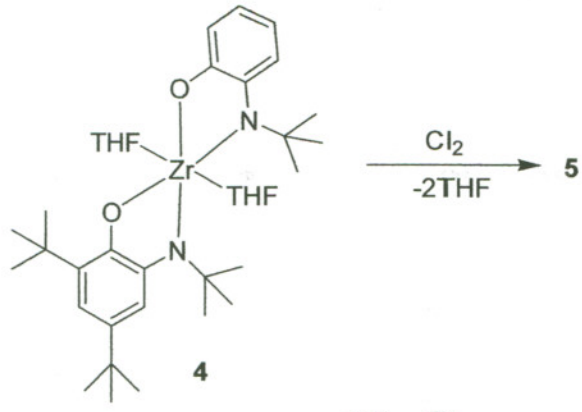
b) $t_{1/2}$ for the thermolysis of **2** in toluene is independent of the concentration of **2**.

c) $t_{1/2}$ for the thermolysis of **2** in toluene is independent of the presence of excess phosphine.

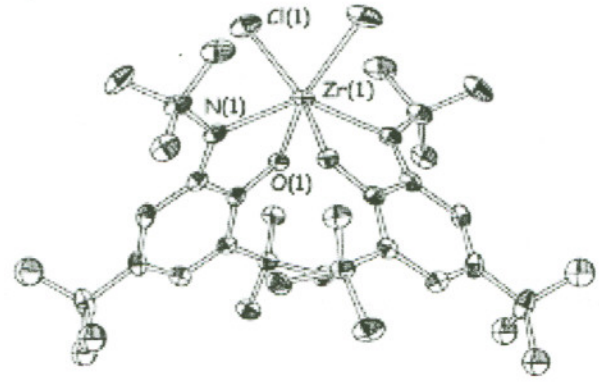
d) Replacing the bromide with iodide led to complete disappearance of the corresponding olefin adduct within two days at room temperature, whereas complex **2** is stable under those reaction conditions.

Given the above kinetic data, **propose a complete step-by-step mechanism** for the conversion of **2** → **3**.

3) Oxidative addition is a fundamental reaction that is central to many transformations in organic chemistry. Oxidative additions proceed by many mechanisms, but both the coordination number and the oxidation state of the resulting metal complex always increase by two units. Consider the reaction recently reported by the Heyduk group at UC Irvine (*Inorg. Chem.* **2005**, 5559) shown in eq 3. Compound **4** was found to react with 1 mol of Cl_2 to give compound **5**. A diagram of the X-ray structure of **5** is shown below.



(3)



3) What is the oxidation state of compound **4**, and how many valence *d*-electrons does the molecule have?

4) Based on your answer in question 3, draw a proper description of compound **5**.